

Urine Dipstick Analyzer v2.0 - Bill of Materials

ESP32-S3 | ILI9341 TFT | OV2640 Camera | DS3231 RTC | MAX17048 Battery Gauge

Category	Ref	Component	Description	Qty	Unit \$	Total \$	Source	
ELECTRONICS - MCU								
	U3	ESP32-S3-DevKitC-1-N8R8	Main MCU with 8MB PSRAM for image processing	1	\$15.00	\$15.00	DigiKey	Link
ELECTRONICS - DISPLAY								
	U10	ILI9341 2.4" TFT Display (320x...	Color TFT for results and UI (replaces SSD1..	1	\$12.00	\$12.00	Amazon	Link
ELECTRONICS - POWER								
	BT1	3.7V 1000mAh LiPo Battery	Rechargeable power source for portable oper..	1	\$10.00	\$10.00	Amazon	Link
	U1	MCP73833 USB-C Charger Module	Lithium charger with USB-C input	1	\$12.50	\$12.50	Adafruit	Link
	U2	Pololu S13V25F9 (9V Regulator)	Step-up/step-down regulator for heater/motor	1	\$12.00	\$12.00	Pololu	Link
	U4	Pololu S7V8F3 (3.3V Regulator)	Efficient 3.3V regulator for MCU and sensors	1	\$6.00	\$6.00	Pololu	Link
	U12	MAX17048 LiPo Fuel Gauge	I2C battery state-of-charge monitor	1	\$5.50	\$5.50	DigiKey	Link
	PCB1	Custom Power Distribution PCB	Power hub with fuse and filtering	1	\$5.00	\$5.00	OSH Park	Link
ELECTRONICS - SENSORS								
	U5	SHT31-D Temperature & Humidity..	High-accuracy ambient environment sensor (l..	1	\$12.00	\$12.00	Adafruit	Link
	U6	BH1750 Ambient Light Sensor	Light level monitoring for ambient rejectio..	1	\$4.50	\$4.50	Amazon	Link
	U11	DS3231 RTC Module	Real-time clock with battery backup for tim..	1	\$6.00	\$6.00	Amazon	Link
	R_NTC	10k NTC Thermistor (B3950)	Tray temperature sensor for PID heater cont..	1	\$0.50	\$0.50	Amazon	Link
	SW3	MC-38 Magnetic Reed Switch	Safety interlock for UV tray detection	1	\$4.00	\$4.00	Amazon	Link
	SW2	Micro Limit Switch (KVV-5)	Tray home position reference	1	\$2.00	\$2.00	Amazon	Link
ELECTRONICS - CAMERA & OPTICS								
	CAM1	OV2640 Camera with CPL Filter	Camera module with circular polarizer for w..	1	\$22.00	\$22.00	Amazon	Link
	U9	GM65 Barcode Scanner Module	UART barcode/QR reader for patient ID scann..	1	\$28.00	\$28.00	Amazon	Link
ELECTRONICS - ACTUATORS								
	D1	5V White LED Array Module	Uniform illumination for strip imaging	1	\$3.00	\$3.00	Amazon	Link
	D2	Common Anode RGB LED	System status indicator (7 colors)	1	\$0.50	\$0.50	Amazon	Link
	D3	275nm UV-C LED Module	Tray sterilization after each test cycle	1	\$15.00	\$15.00	DigiKey	Link
	M1	NEMA 17 Stepper Motor (20mm bo..	Drives dipstick tray to precise focal posit..	1	\$12.00	\$12.00	Amazon	Link
	U7	A4988 Stepper Motor Driver	Step/direction motor controller with micros..	1	\$5.00	\$5.00	Amazon	Link
	HTR1	Polyimide Heating Film (25x50mm)	Flexible heater for 37°C enzyme incubation	1	\$8.00	\$8.00	Amazon	Link
	FAN1	30mm 5V Brushless DC Fan	Post-measurement venting and dehumidification	1	\$5.00	\$5.00	Amazon	Link
	BZ1	5V Active Piezo Buzzer	Audible alerts for results and errors	1	\$1.50	\$1.50	Amazon	Link
ELECTRONICS - SWITCHING								
	Q1	2N7000 N-Channel MOSFET	Fan switching (low-side)	1	\$0.50	\$0.50	Amazon	Link
	Q2	IRLZ44N Logic Level MOSFET	UV-C LED driver (high current)	1	\$1.00	\$1.00	Amazon	Link

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	Q3	IRLZ44N Logic Level MOSFET	Heater control (high current PWM)	1	\$1.00	\$1.00	Amazon	Link
ELECTRONICS - PASSIVE & MISC								
	SW1	Tactile Push Button 6x6mm	Measure/navigate button	1	\$0.50	\$0.50	Amazon	Link
	SD1	Micro SD Card Module (SPI)	Data logging and raw image storage	1	\$4.00	\$4.00	Amazon	Link
MECHANICAL - ENCLOSURE								
	ENC1	Injection-Molded Enclosure Base	180x120x40mm base with mounting bosses and ..	1	\$25.00	\$25.00	Custom Mold	
	ENC2	Injection-Molded Enclosure Cover	Matching cover with display window and snap..	1	\$7.00	\$7.00	Custom Mold / 3D Print	
	TRAY1	Injection-Molded Dipstick Tray	120x25x8mm tray with strip channel and T8 n..	1	\$10.00	\$10.00	Custom Mold / 3D	
MECHANICAL - MOUNTS & BRACKETS								
	MNT1	Camera & LED Mount Bracket	3D-printed bracket for camera and LED at op..	1	\$4.00	\$4.00	3D Print	Link
	MNT2	Barcode Scanner Mount	3D-printed bracket for GM65 alignment	1	\$3.00	\$3.00	3D Print	Link
	MNT3	Exhaust Vent Grill	3D-printed slotted grill for fan port	1	\$2.00	\$2.00	3D Print	
	MNT4	Internal Cable Management Frame	Wire routing skeleton to keep optics clear	1	\$2.00	\$2.00	3D Print	
MECHANICAL - MOTION								
	LS1	T8 Lead Screw 100mm + Copper Nut	Linear motion conversion for tray positioning	1	\$8.00	\$8.00	Amazon	Link
MECHANICAL - FASTENERS & SEALS								
	HW1	M2x6mm Pan Head Screws	PCB and module mounting	12	\$0.05	\$0.60	Amazon	Link
	HW2	M3x8mm Pan Head Screws	Cover attachment	4	\$0.10	\$0.40	Amazon	Link
	HW3	M3 Brass Heat-Set Inserts	Reusable threads in 3D-printed/molded parts	4	\$0.20	\$0.80	Amazon	Link
	HW4	M2.5 Brass Hex Standoffs	Electronics mounting	8	\$0.50	\$4.00	Amazon	Link
	GSK1	Silicone Gasket Seal	Watertight seal between base and cover	1	\$3.00	\$3.00	Amazon	Link
	GSK2	EPDM Foam Strips	UV-light-tight gasket for enclosure seams	1	\$5.00	\$5.00	Amazon	Link
MECHANICAL - OPTICAL								
	OPT1	Frosted Acrylic Diffuser (30x2..	LED light diffusion panel	1	\$2.00	\$2.00	Amazon	Link
	OPT2	Fused Quartz Window (20x20mm)	UV-transparent shield over camera	1	\$8.00	\$8.00	Amazon	Link
	OPT3	Telescope Flocking Paper	Anti-reflection coating for tray interior	1	\$10.00	\$10.00	Amazon	Link
MECHANICAL - CALIBRATION								
	CAL1	Mini Color Calibration Card	Fixed reference for firmware color calibrat..	1	\$1.00	\$1.00	Amazon	Link
	CAL2	18% Gray / White Balance Card	White balance and exposure reference target	1	\$5.00	\$5.00	Amazon	Link

TOTAL (72 parts)						\$314.80		
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Electronics subtotal: \$214.00
Mechanical subtotal: \$100.80

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Notes: Prices are approximate and based on single-unit purchases. Volume pricing (100+ units) typically reduces costs by 30-50%. The ILI9341 TFT, DS3231 RTC, and MAX17048 fuel gauge are v2.0 upgrades replacing the original SSD1306 OLED. Custom injection-molded parts (enclosure, tray) can be 3D-printed for prototyping.